Krishiv Singhal

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Education

Carnegie Mellon University *Aug 2024–Dec 2027*
Major in Mathematics and Logic and Computation *United States*

- Notable coursework: Advanced NLP (PhD), Machine Learning (PhD), Cooperative AI (Graduate), Modal Logic (Graduate), Formal Logic, Probability Theory, Linear Algebra, Abstract Algebra, Real and Complex Analysis. GPA: 3.7.
- Studied abroad at the University of Sydney; completed advanced coursework in maths on a full merit scholarship; awarded the Dalyell Scholarship and Chancellor's Award. GPA: Distinction.

United World College, Singapore *Aug 2017–May 2024*
International Baccalaureate *Singapore*

- Perfect IB Score: 45/45 (top 0.5% worldwide); ranked first in class for thesis on the Curry–Howard correspondence and intuitionism.

Experience

Quantitative Research Intern *Jun–Aug 2026*
Stealth Systematic Macro Fund; reference available on request *Hong Kong*

- Built Bayesian state-estimation engine that classified global equity indices and stocks into market regimes using Gaussian-kernel geometry; integrated into the fund's research stack.
- Developed co-activity and flow measures tracking probability mass between regimes, isolating churn conditions.

Modal Logic and Reinforcement Learning *Mar 2026–Present*
Research, Carnegie Mellon University

- Formulated novel probabilistic semantics for propositional dynamic logic by relating programs to Markov decision processes; researching its use as a framework for reinforcement learning.

AI Evaluation Research Group *Sep 2026–Present*
Carnegie Mellon University

- Selected to research prediction and recalibration of model performance under covariate shift, including synthetic distribution-shift experiments and unlabeled recalibration methods.

Computational Linguistics Research Group *Sep 2024–Feb 2025*
Carnegie Mellon University

- Analyzed human and AI generated writing through eigendecomposition of stochastic matrices; found statistically significant variations in Markov Chain mixing time(s).

Projects

Llama-Style Language Model *PyTorch*

- Implemented a 25M-parameter Llama-style transformer model in PyTorch, including BPE, rotary embeddings, grouped-query attention, RMSNorm, SwiGLU, and AdamW; trained on a custom 212M-token literary and philosophical corpus, with controlled upsampling experiments across selected authors.

Differentiable Kripke Semantics for Modal Model Search *PyTorch, NumPy*

- Implemented differentiable fuzzy Kripke semantics and cast modal satisfiability as gradient descent on the accessibility relation, posed as a minimax against an adversarial valuation. Developing a framework that uses modal constraints as a regulariser for RL policies.

Bitewise Labs — Bites Suite *Flutter, Swift*

- Founded an iOS studio with three shipped cultural-education apps (Literature, Art and Architecture Bites) with thousands of impressions and consistent revenue. Built automated video-generation and agent-led marketing content pipelines.

Finite-Model Theory / FO² Decision Problem — Invited Seminar Presentation

- Researched, prepared and delivered a talk to faculty and members of the Australasian logic seminar on the decision problem for the binary fragment of first-order logic.

Awards

Chancellor's Scholarship, the highest undergraduate award at the University of Sydney · **UWC Global Academy,** fully funded selective residential program at the University of Chicago · **Canadian Senior Maths Contest,** International Honour Roll, global top 1% · **AIME,** Qualifier, global top 2.5%.

Skills and Interests

Languages: Python, SQL, LaTeX · **Machine learning:** PyTorch, NumPy, pandas · **Also:** Flutter, Lean 4.

Interests: Music production (composed, produced, and released 5 albums on major streaming platforms; remix favourite songs on YouTube); golf; poker.